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Post Jotnian basic Intrusions in the Fennoscandian Shield, and the break up of Baltica from Laurentia: a palaeomagnetic and AMS study

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Abstract

A palaeomagnetic and anisotropy of magnetic susceptibility (AMS) study has been performed on dolerite sills of the Central Scandinavian Dolerite Group (CSDG) in the Fennoscandian Shield. The dolerites occur in four previously known complexes in central Sweden and Finland and from the results of this palaeomagnetic study another complex has been identified in northern Sweden. These complexes cover an area of at least 100 000 km^2 and the palaeomagnetic data suggest a small difference in time between the intrusion of the dolerites. The measurements of anisotropy of magnetic susceptibility reveal a magnetic fabric with almost horizontal foliation planes and lineations that indicate fairly uniform ca NW or SE directed magma flows. The dolerites of the CSDG are geochemically rather uniform and have compositions typical of mantle derived melts formed in continental tensional settings. In a palaeomagnetic reconstruction of Baltica versus Laurentia at ca 1.27 Ga the two continents were joined, with NE Greenland attached to NW Baltica. AMS data from a few dolerites and a basalt in NE Greenland indicate magma flow directions that in the tectonic reconstruction are more or less parallel to the flow of the dolerites in Sweden. This may suggest a common magma source located at the reconstructed contact between Baltica and Laurentia. Both the dolerites in Greenland and those in Sweden are of tholeitic composition indicating an intraplate origin, which supports the interpretation of joined continents at that time. The tensional regime, that is reflected by the huge sill complexes, is in our interpretation related to the break up of Baltica from Laurentia at ca 1.27 Ga ago. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Baltica; Dolerite sills; Palaeomagnetism; Magnetic susceptibility; Geochemistry

1. Introduction

Mafic dykes represent an expression of crustal extension and are the feeding channels for basaltic magma when it is transferred from the mantle to the upper crust. On the continents mafic dykes have been intruded throughout the Proterozoic

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and Phanerozoic. Most late Phanerozoic dyke swarms are associated with divergent plate margins, however, some have been emplaced as a result of plate subduction (e.g. Nakamura, 1977), back arc spreading (Church, 1985) and collision (Feraud et al., 1987). In the Proterozoic the association between dyke intrusions and plate tectonic processes is not that clear as plate boundaries have been affected by later collisional events. Metamorphism and erosion of the crust make products of Proterozoic passive margin events difficult to trace and basic dykes of this age can therefore be more difficult to put into a plate tectonic context.

In the Fennoscandian Shield various generations of dyke complexes intruded the shield during the Precambrian. At ca 1.27 Ga mafic gently dipping sills and dykes (Welin and Lundqvist, 1975; Patchett, 1978; Suominen, 1991), the Central Scandinavian Dolerite Group (CSDG; Gorbatschev et al., 1979), intruded the central part of Sweden and southwestern part of Finland, extending over an area of at least ca 100 000 km^2. This group of mafic intrusions is coeval with large igneous events in the North Atlantic area, e.g. the Gardar dykes in Greenland and the Mackenzie and Sudbury dykes in Canada. The purpose of this study is to put these Mesoproterozoic Fennoscandian sill complexes into a plate tectonic framework.

Basic dykes have proven to be excellent palaeomagnetic recorders and consistent results have also been reported from these sills and dykes in Finland and Sweden (Neuvonen, 1965; Neuvonen and Grundström, 1969; Poorter, 1976; Piper, 1979; Magnusson, 1983; Bylund, 1985). By calculating and comparing the palaeomagnetic poles for mafic intrusions in various parts of the shield we want to define the extension of the CSDG and we also look for eventual differences in time of intrusion of the dolerites. Geochemical analyses of the sills will provide information of the geological environment at the time of intrusion and may give constraints to plate tectonic reconstructions. By measurement of anisotropy of magnetic susceptibility (AMS) preferred orientation of magnetic grains in the sills can be defined. In basic dykes AMS measurements have shown that there is a relation between direction of magma flow and orientation of grains (Knight and Walker, 1988). Here we measure the AMS of the sills in an attempt to define the flow direction of the magma and look for eventual geographical variations.

2. Geology of the Fennoscandian Shield and 1.25 Ga dyke intrusions in Greenland and North America

The Fennoscandian Shield is divided into a number of tectono-magmatic blocks (Fig. 1), with the oldest parts in the northeast. These blocks represent a geochronological zonation and an accretionary growth of Fennoscandia during several successive orogenic events, i.e. the Loopian (2.9–2.6 Ga), the Svecofennian (1.9–1.85 Ga), the Gothian, and the Trans-Scandinavian Granite and Porphyry Belt (1.81–1.5 Ga), the Sveconorwegian (1.25–0.85 Ga) and the Caledonian (0.6–0.4 Ga) events, respectively. The Svecofennian province forms the central part of the shield and is divided into northern, central and southern subprovinces (Gaal and Gorbatschev, 1987; Fig. 1). Rocks of the Trans-Scandinavian Granite and Porphyry Belt (TGPB; Gorbatschev, 1985) and the Gothnian orogeny truncate the Svecofennian province in the southwest and the Southwest Scandinavian province, which was formed during the Gothnian orogeny and strongly reworked during the Sveconorwegian orogeny. Large volumes of mafic dykes were intruded within two different time intervals during and after the Gothnian orogeny. At ca 1.65–1.3 Ga dykes associated with anorogenic granite emplacement intruded the crust in the central parts of the shield and at ca 1.27 Ga mafic sills and dykes of the CSDG intruded primarly into the southern and central Svecofennian subprovinces and the TGPB. The Central Scandinavian Dolerites have been divided into four geographically separated complexes, the Satakunta, the Jämtland, the Ulvö and the Dala complexes (Fig. 1). Flat lying mafic intrusions have been identified in drilling and from seismic data in deep parts of the crust beneath the Siljan Impact Structure (Juhlin and Pedersen, 1987) and probably also from seismic data in the basement

beneath the Caledonian nappes in Jämtland (Palm et al., 1991). Seismic reflection data from the Bothnian Bay also reveal several prominent reflectors in the crust which are interpreted as dolerite sills (BABEL Working Group, 1993), suggesting a more or less continuous belt of dolerites from Sweden to Finland. The large areal extent of the sill complexes indicates that a significant volume of mafic magma was added to the Fennoscandian crust. Palaeomagnetic data are consistent within and between the different complexes (Bylund and Pesonen, 1987), which may suggest that there was

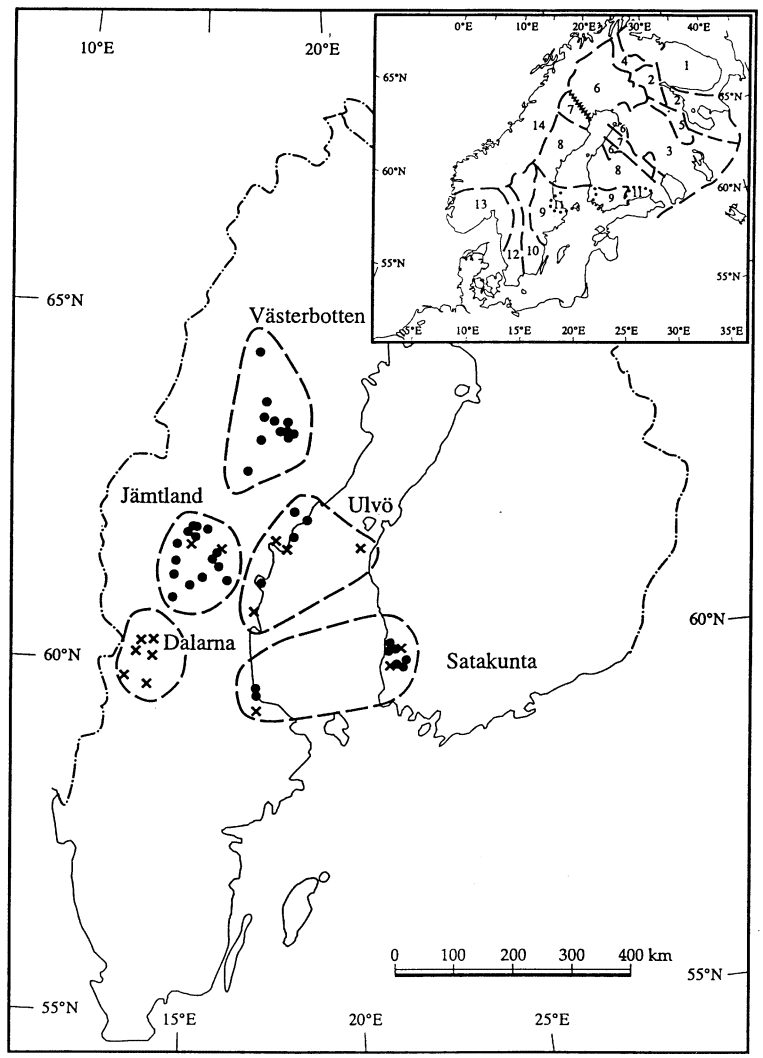


Fig. 1. The geographical distribution of sampled sites of this study (dots) and palaeomagnetic sites from previous investigations (crosses). The various Postjotnian sill complexes are defined by the hatched lines. In the inset the tectonomagmatic block division of the Fennoscandian Shield (modified after Gaal and Gorbatschev, 1987; Elming et al., 1993) is shown. The numbers correspond to the following blocks: *Archaean* province: (1) Kola; (2) Belomoria; and (3) Karelia subprovinces: *Svecofennian* province: (4) Lappland granulite belt; (5) Jatulian/Svecofennian outlier in Karelia; (6) Northern Svecofennian subprovince; (7) Skellefte and Raahe Ladoga belts; (8) Central Svecofennian subprovince; (9) Southern Svecofennian subprovince; (10) Transscandinavian Granite-Porphyry Belt; (11) Rapakivi granites: *Southwestern Scandinavian* province: (12) Protogine zone; (13) Southwest Scandinavian subprovince; (14) *Caledonides*.

little or no difference in time between the intrusions.

Dyke swarms and sills of similar age are found in the southern and northern parts of Greenland. In south Greenland, in the 1.9–1.7 Ga old Ketilidian mobile belt, rifting took place during the Gardar period (1.3–1.1 Ga; Emeleus and Upton, 1976). This rifting is characterized by intrusions of alkaline complexes and swarms of WNW to NE trending basic to salic dykes emplaced in a zone some 200 km long and 70 km wide. Four generations of dykes have been recognized (BD0–BD3; Berthelsen and Henriksen, 1975) on the basis of their trends. Dykes belonging to the W–E trending BD1 group have been dated at 1.249 ± 0.029 Ga (at Ivigtut) and 1.228 ± 29 Ga (at Arsuk; Rb–Sr, whole rock; Patchett et al., 1978), which is very close to the age of the CSDG. Palaeomagnetic data have been reported from the different generations of dykes, from alkaline complexes, lavas and sedimentary rocks (Piper, 1992; Thomas and Piper, 1992; Piper, 1994, 1995; Piper et al., 1998), representing ages of magnetization from ca 1.30 to 1.16 Ga. At about 1.25 Ga intense volcanic activity took place in eastern north Greenland. This resulted in large sill complexes and a sequence of tholeitic flood basalts, the Zig–Zag Dal Basalts (Kalsbeek and Jepsen, 1983). The dolerite sills and dykes have a quartz–tholeitic composition (Jepsen, 1971; Jepsen and Kalsbeek, 1979) and intrude Middle Proterozoic sandstones of the Independence Fjord Group. Hydrothermal alterations in intrusions related to the sill complexes have Rb–Sr isochron ages of ca 1.23 Ga (Jepsen and Kalsbeek, 1979; Kalsbeek and Jepsen, 1983). These alterations probably occured not long after the time of intrusion of the sills (Kalsbeek and Jepsen, 1983). Palaeomagnetic data and some AMS data have been reported from the basalt and the sill complexes (Marcussen and Abrahamsen, 1983).

In the Laurentian Shield the Sudbury and Mackenzie dykes are of similar age (1.267 ± 0.002 Ga; LeCheminant and Heaman, 1989 and $1.235 \pm 0.007/-3$; Dudàs et al., 1994, respectively) as the CSDG in the Fennoscandian Shield. The Mackenzie dykes form a huge swarm of N–NW trending dykes in the Churchill Province and geochemical data (Fahrig and West, 1986; Gibson et al., 1987) suggest that they are not related with the WNW trending Sudbury dykes in the southern Superior Province.

3. Sampling

Samples for palaeomagnetic and AMS analyses were collected from the CSDG complexes in Jämtland, Ulvön and Satakunta areas. Samples of sills and dykes were also taken from an area north of ca 63° latitude in Sweden, where members of the Central Scandinavian Dolerite Suite have so far not been identified (Fig. 1). Geochemical analyses were made on the samples from this area and on the samples from Jämtland.

The samples were collected with a portable drill and as block samples and were oriented using magnetic and sun compasses. Generally each site represents at least 20 m^2 of outcrop. The study includes 336 samples collected at 39 sites distributed over an area from 61.0° latitude and 22.17° longitude in the southeast (southwestern Finland) to 65.73° latitude and 17.32° longitude in the northwest (northern Sweden).

4. Palaeomagnetic results

The specimens were demagnetized by conventional alternating field and thermal demagnetization techniques and the remanent magnetizations were measured with a spinner (Czech-JR5) and a SQUID (2G-DC) magnetometer. The different components of remanence were defined by least square fitting of vectors in three-dimensional space (Torsvik, 1986; Torsvik et al., 1996) and by principal component analyses (Kirschvink, 1980). The vectors were generally defined with mean angular deviations smaller than 7°.

It was possible to define a characteristic magnetization in 38 of the 39 sampled sites.

4.1. The Jämtland complex

Dolerite sills in the Jämtland area were sampled in 15 geographically well separated sites. From

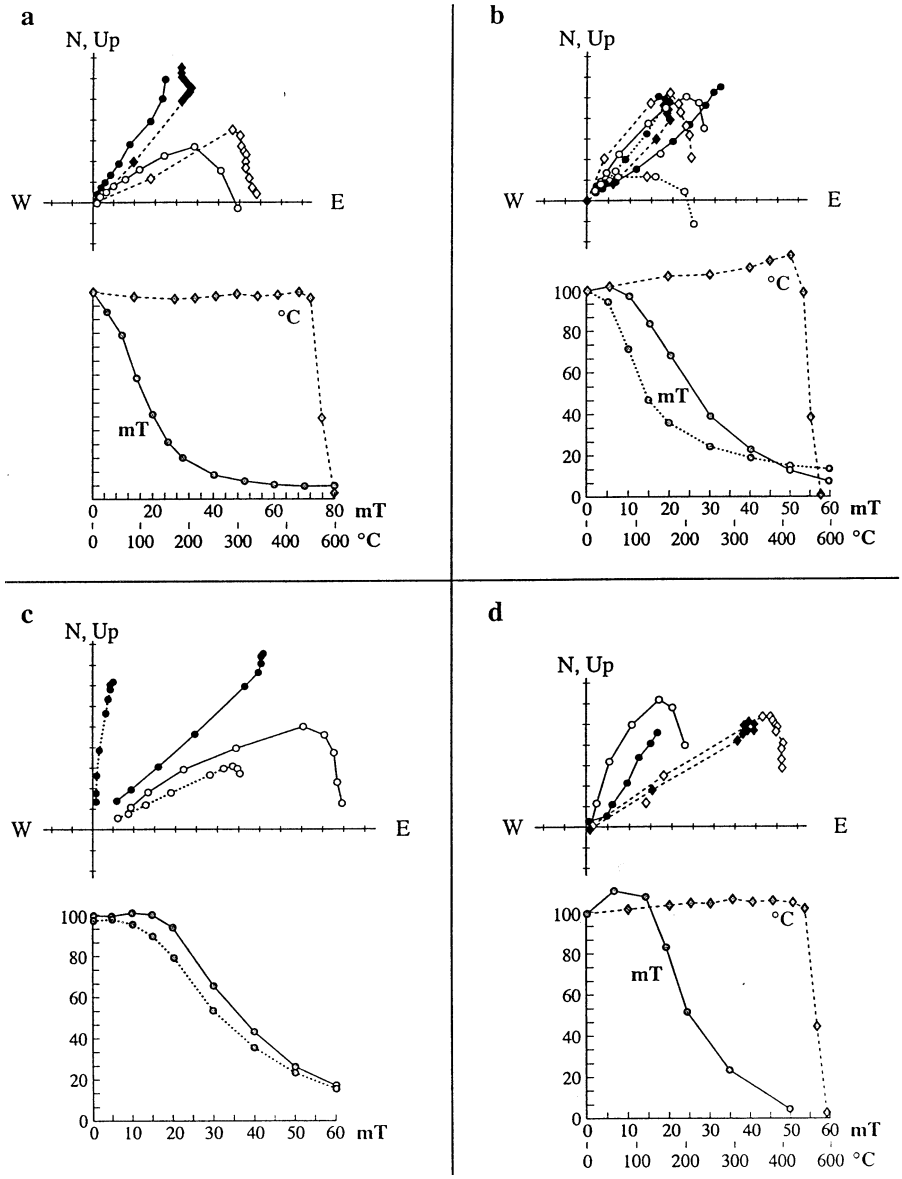


Fig. 2. Examples of remanence behaviour during progressive alternating field and thermal demagnetization is shown in vector plots of dolerite samples from: (a) the Jämtland Complex; (b) the Ulvö Complex; (c) the Satakunta Complex; and (d) sills in northern Sweden (the Västerbotten Complex). Open (closed) symbols in the vector plots denote projections in a vertical (horizontal) plane.

this complex only two sills have been studied before (Bylund, 1985; Fig. 1Table 2). The dolerites often have an ophitic texture and carry large amounts of magnetite surrounded by ilmenite lamellae. Plagioclase, biotite, titan-rich augite and olivine are common minerals and minor amounts of pyrite, pyrrhotite and chalcopyrite are often present. The sills are classified as olivine diabases. The Koenigsbereger-ratio (Q-value) varies, and is as high as 30 at some sites, however at the majority it is in the range 0.62–1.0. In 14 of the sites it was possible to define a characteristic magnetiza-

tion usually isolated in a narrow range of unblocking temperatures of ca 560–580°C (Fig. 2a). The coercivity of the natural remanent magnetization (NRM) is generally high, with median destructive fields (MDF; Dunlop and Özdemir, 1997) of ca 20 mT (Fig. 2a). This suggests that the carrier of remanence is predominantly pseudo-single-domain magnetite-titanomagnetite. There is a small portion of soft remanence, which is usually erased at 15–20 mT demagnetizing fields. This soft remanence carries a magnetization with a direction similar to that of the present earth's field (PEF; Fig. 2a). Also in thermal demagnetizations a low unblocking temperature component is isolated and from the similarity in direction with that of the PEF a viscous origin for this remanence is suggested (Fig. 2a).

The site mean directions are fairly well defined with precision parameters (Fisher, 1958) generally higher than 40 (Table 1). The site mean directions are distributed within a narrow range of declinations but wide range of inclinations (Fig. 3a). This distribution of directions is sometimes also seen during alternating field demagnetization of individual samples, where components of higher negative inclinations are isolated in higher coercivity ranges. The reason for the gradual changes in direction is probably due to overlapping coercivity spectra, between the viscous and the primary magnetization. Such gradual changes in directions are also observed during thermal demagnetization, but here the two components are clearly separated (Fig. 2a). This means that imperfect cleaning and isolation of the different components of magnetization during AF demagnetizations may explain the distribution of site mean directions. At site 29 the sill has baked the host rock, the Rätan granite (Mattsson and Elming, 2000). In other localities in that region a characteristic magnetization has been defined for the granite, with a direction of magnetization significantly different from that of the sill. This suggests that the high unblocking temperature and coercivity magnetization in the sills relate to a primary origin. These site mean directions form a rather well defined mean direction of characteristic magnetization for the Jämtland Complex at a declination of 37.8° and inclination of −37.3° ($\alpha_{95} = 9.1°$; Fig. 4 Table 3).

4.2. The Ulvö complex

Sills within the Ulvö Complex have been sampled at four sites. The sill at Ulvön is usually medium grained and layered. The layering is expressed by prefered orientation of tabular plagioclase, and to some extent augite crystals, subparallel with the plane of layering (Lundqvist and Samuelsson, 1973; Larsson, 1980). The main minerals are plagioclase, augite, olivine, biotite and Fe–Ti oxides. This complex has earlier been subject to extensive palaeomagnetic studies (Neuvonen, 1966; Poorter, 1975; Magnusson and Larsson, 1977; Piper, 1979, 1980; Table 2). Three of the sites in our study are located at the northern and southern edges of the complex and only one is from the more central part (Fig. 1).

The magnetic properties of these sills are similar to those of the Jämtland Complex. The coercivity of remanence varies from medium high to high (MDF's of 12–26 mT; Fig. 2b) and also in these sills a low coercivity component carries a direction close to that of PEF. The unblocking temperature of the characteristic magnetization is in the range 550–580°C and in the thermal demagnetizations a low unblocking temperature component with a PEF direction can be isolated (Fig. 2b). At site 15 the gradual change in directions during thermal demagnetization falls along great circles that intersect (decl. = 67.1, incl. = −66.3) not very far from the mean direction defined from the high unblocking temperature components (decl. = 49.1, incl. = −51.7; Table 1). The sill at this site has probably baked a rapakivi granite, while dykes related to the rapakivi intrusion are not affected (Moakhar and Elming, 2000). This may indicate a primary origin of the remanent magnetization. The site mean directions are most often very well defined (Fig. 3b) and they cluster around a mean for the Ulvö Complex with a declination of 48.1° and inclination of −45.3° (Fig. 4; Table 3).

4.3. Satakunta complex

The sill complex in southwestern Finland cuts Svecofennian and rapakivi rocks and Jotnian sandstone. The dolerites have a subophitic texture

Table 1
New palaeomagnetic data from Post-Jotnian dolerites in the Fennoscandian Shield[a]

Site	Glat/Glon.	*n*	Decl.	Incl.	*k*	α_{95}	Plat.	Plon.	A_{95}
Northern Sweden									
(1) Fräkenmyrberget	65.73/17.32	8	34.9	−33.7	84	6.1	1.8	166.4	5.3
(2) Storgranliden	65.00/17.62	8	29.6	−43.3	48	8.1	−3.1	171.0	8.0
(3) Middagsberget	64.77/17.55	6	44.9	−41.5	236	4.4	−5.2	157.1	4.2
(4) Kvarnberget	64.72/17.90	8	55.6	−31.8	17	14 1	−2.1	145.8	11.9
(5) Umgransele	64.70/18.40	6	46.7	−39.1	65	8.4	−3.9	155 9	7.7
(6) Orrkulla	64.57/18.42	13	42.7	−49.6	39	6.7	−10.7	161.9	7.2
(7) Svanamyrliden	64.55/18.17	6	55.7	−59.5	40	10.7	−23.7	154.8	14.0
(8) Björnliden	64.52/18.57	7	56.5	−52.8	341	3.3	−17.4	152.7	3.8
(9) Örån	64.50/18.45	6	40.3	−56.7	37	11.1	−16.6	166.0	13.7
(10) Gammrenberget	64.43/17.47	5	46.0	−38.5	32	13 8	−3.1	155.4	12.6
(11) Åkerberget	63.97/17.01	6	15.0	−20.6	54	3.3	−14.5	181.8	7.0
Ulvö complex									
(12) Skagsudde	63.23/19.02	10	55.1	−38.3	216	3.3	−5.1	149.0	8.4
(13) Stybbersmark	63.36/18.58	9	48.7	−52.4	117	4.8	−13.8	158.1	5.4
(14) U1vön	62.98/18.62	6	39.7	−38.3	54	9.2	−0.1	162.2	8.4
(15) Gubben	62.35/17.57	12	49.1	−51.7	178	3.3	−12.5	156.7	3.7
Jämtland complex									
(16) V. Stugun	63.10/15.54	8	39.9	−36.1	53	7.7	1.2	158.5	6.9
(17) NO Östersund (a)	63.10/15.19	7	39.5	−44.9	125	5.4	−4.9	160.3	5.4
(18) NO Östersund (b)	63.09/15.07	8	48.7	−29.0	29	10.5	2.8	148.7	8.6
(19) N. Pilgrimstad	63.02/14.95	6	39.8	−37.5	90	7.1	0.4	158.2	6.4
(20) Hackås	62.85/14.53	15	29.0	−26.0	43	6.0	10.2	165.9	4.8
(21) Övertjärnen	62.64/15.78	7	37.2	−28.9	22	13.1	6.7	159.8	10.7
(22) Borgsjöby	62 64/15.94	5	46.8	−58.7	41	12.1	−18.7	159.5	15.5
(23) Ö. Nästelsjön	62.59/14.50	10	46.0	−18.0	43	8.0	10.0	148.4	6.0
(24) Ljungaverken	62.51/16.03	6	44.7	−59.2	50	9.6	−18.6	161.4	12.4
(25) Rätansbyn	62.40/14.50	10	31.0	−19.0	38	8.0	13.9	163.0	6.0
(26) Ånge	62.38/15.47	8	61.2	−73.3	110	5.3	−40.2	159.3	9.0
(27) Munkbyn	62.33/16.31	6	32.2	−40.1	23	14.4	1.1	166.9	13.5
(28) W. Havern	62.24/15.09	10	28.0	−16.0	54	7.0	16.3	166.1	5.2
(29) Ytterberg	62.06/14.50	16	28.0	−34.0	77	4.0	6.3	167.9	3.4
Satakunta complex									
(30) Gammelhällan	60.75/17.36	6	24.7	−16.9	98	6.8	17.9	171.6	5.0
(31) Skutskär	60.65/17.39	13	11.3	−19.2	125	3.7	18.9	185.6	2.8
(32) Lappi	61.10/21.90	10	39.5	−40.8	77	5.5	−0.3	166.2	5.2
(33) S. Eura	61.05/22.10	10	46.3	−37.6	297	2.8	−0.2	159.5	2.5
(34) E. Eura	61.13/22.18	10	53.2	−36.6	103	4.8	−1.9	153.5	4.3
(35) NW. Panelia	61.28/21.90	10	41.8	−48.8	43	7.4	−7.1	166.2	7.9
(36) N. Luria	61.41/21.67	12	47.4	−44.5	99	4.4	−5.5	160.1	4.4
(37) S. Luria	61.30/21.67	10	42.8	−36.4	139	4.1	1.6	162.0	3.7
(38) W. Nakkila	61.33/21.83	7	41.4	−36.7	119	5.6	1.8	163.5	5.6

[a] *Note:* Glat/Glon, geographical latitude and longitude of the site; *n*, number of samples; Decl., mean declination; Incl., mean inclination; *k*, Fisher precision parameter; α_{95}, radius of 95% confidence circle; Plat., Plon., latitude and longitude of calculated palaeomagnetic pole positions; A_{95}, half-angle of 95% circl of confidence.

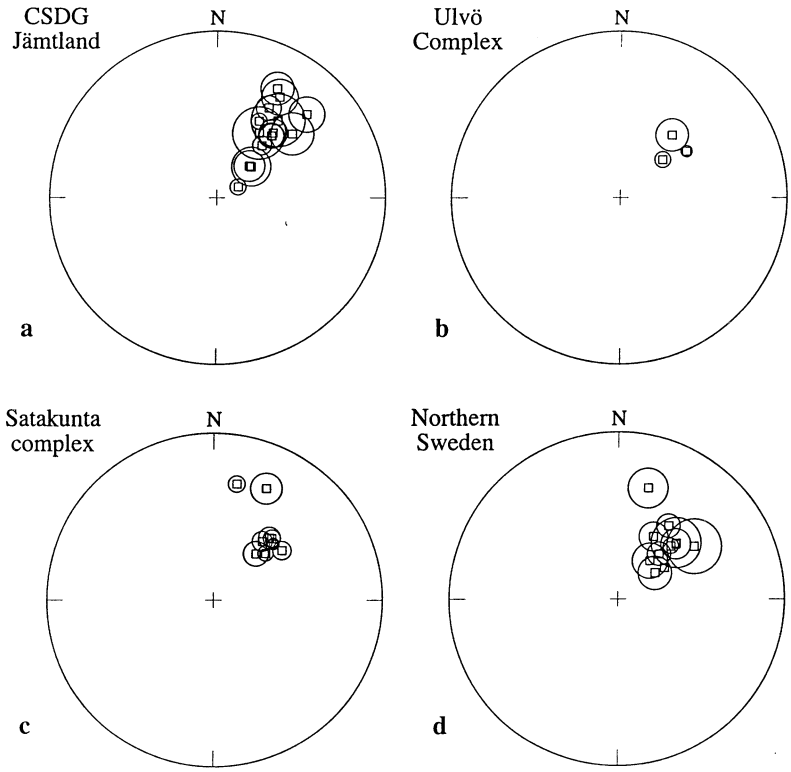


Fig. 3. Site mean directions of the characteristic magnetization for: (a) the Jämtland Complex; (b) the Ulvö Complex; (c) the Satakunta Complex; and (d) sills in northern Sweden (the Västerbotten Complex).

and the main minerals are plagioclase, augite, olivine and ilmenomagnetite (Hämäläinen, 1987). Samples of the dolerites were collected in seven sites (Fig. 1). The coercivity of remanence is high for these sills with MDFs of 30–40 mT and the unblocking temperatures are restricted to a narrow temperature range between 560 and 580°C indicating that the carrier of remanence is pseudo-single-domain magnetite-titanomagnetite. Also in these sills the direction of remanence changes from low negative to intermediate high inclinations during AF demagnetization. A stable characteristic magnetization is defined after demagnetization in fields higher than 40 mT, with a vector declining towards the origin in the vector plot (Fig. 2c). The intensity of remanence increases during the first steps of demagnetization and in the very low coercivity range it is possible to isolate a remanence component with a direction of positive inclination not far from that of the PEF (Fig. 2c). This explains the gradual change of direction of the remanence vector during progressive demagnetization, similar to what has been observed for sills in other parts of the shield (e.g. in the Section 4.1).

The mean directions of characteristic magnetizations at the different sites are well defined with precission parameters (Fisher, 1958) higher than 43 (Table 1) and they form a tight cluster around a mean direction for the Finnish sills at 44.7° declination and $-40.3°$ inclination (Fig. 4; Table 3).

On the Swedish side of the Bothnian Bay a basic dyke and a sill (sites 30 and 31; Fig. 1) probably belonging to the CSDG were studied. The MDFs are high for both of them varying between ca 25 and 35 mT and characteristic magnetizations were isolated in demagnetizing fields of 30 mT and higher (Fig. 2c). Also in these dolerites a soft remanence with a direction similar

Table 2
Previous palaeomagnetic data from central Scandinavian dolerites[a]

Dyke swarm	Glat./Glon.	B/n[b]	Decl.	Incl.	α_{95}	K	Plat.	Plon.	A_{95}	Ref
Ulvö complex										
Mean Ulvö dolerites	63.6/18.5	2/52	45.1	−37.4			−1.5	157.0		Magnusson and Larsson, 1977; Larsson and Magnusson, 1976
Vasa dolerite	63.0/20.9	15/108	38.0	−29.0	5.0	54	7.0	164.0	4.2	Neuvonen, 1966
Dykes										
V. Norrland dolerites	62.9/18.3	43/252	47.8	−48.5	2.6	71	−7.5	156.5	2.6	Piper, 1979
Mean Nordingrå	62.8/18.4	2/21	38.4	−38.2			0.4	163.1		Fridh, 1979; Piper, 1980
Dolerites										
Gnarp sills	62.1/17.3	2/12	59.8	−47.7			−12.7	146.4		Poorter, 1975
Gnarp dolerites	62.1/17.2	1/6	45.8	−51.8	8.0	78	−11.0	159.0	8.4	Poorter, 1975
Mean Ulvö complex		6/451	45.2	−41.8	8.5	64	−4.1	157.6	8.2	
Jämtland complex										
Sundsjö dolerite	63.0/15.1	5/5	40.2	−52.2	16.7	22	−11.0	161.5	19.0	Bylund, 1985
Gimån dolerite	62.8/16.0	18/18	51.4	−41.2	8.1	19	−5.4	150.0	7.7	Bylund, 1985
Mean Jämtland complex		2/23	46.4	−46.8			−8.1	155.3		
Dala complex										
Jotnian dolerites	61.7/13.1	1/5	37.0	−33.0	4.0	30	5.0	158.0	3.9	Piper and Smith, 1980
Mossi dolerite sill	61.6/13.8	1/5	45.3	−24.0	35.1	6	20.3	151.3	27.6	Bylund, 1985
Älvdalsåsen dolerite sill	61.4/13.8	1/11	49.5	−37.6	5.3	74	1.4	148.6	4.8	Bylund, 1985
Älvho dolerite sill	61.4/14.8	1/5	23.6	−16.4	12.1	41	17.8	170.2	8.9	Bylund, 1985
Särna dolerites	61.0/13.0	5/27	36.5	−33.1	4.4	301	5.3	158.6	3.7	Piper, 1980
Lybergsgnupen	60.8/13.7	5/5	31.0	−42.0	10.0	62	1.3	165.7	9.6	Mulder, 1971
Dolorite										
Mean dala complex (Mossi dolerite sill excluded)		5/53	35.1	−32.7	12.0	42	5.9	160.4	10.2	
Satakunta complex										
Satakunta sills	61.2/22.0	2/9	35.5	−34.0			5.3	168.5		Neuvonen, 1965
Stakunta	61.2/22.0	18/18	46.0	−34.0	4.0	60	2.0	158.0	3.9	Neuvonen, 1965
Dolerite dykes										
Gävle dolerite	60.5/17.1	2/15	48.5	−24.3	4.0	104	7.0	150.0	2.8	Poorter, 1976
Märket dolerites	60.3/19.3	8/58	60.0	−40.0	9.2	36	−5.9	145.5	8.8	Neuvonen and Grundström, 1969
Mean Satakunta complex		4/100	47.3	−33.4	12.0	60	2.2	155.7	10.3	

[a] *Note:* The mean directions are calculated from the site when data from more than one site exists. For other explanations see Table 1.
[b] B, number of formations or sites; n, number of samples.

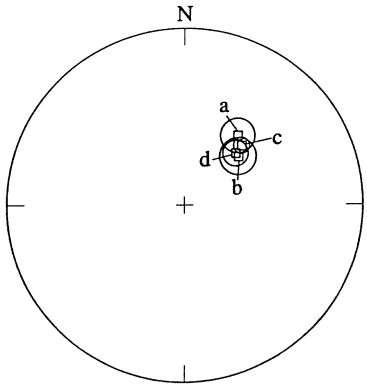


Fig. 4. Formation mean direction of: (a) the Jämtland; (b) the Ulvö; (c) the Satakunta; and (d) the Västerbotten Complex as calculated from the site mean directions of this study.

to that of PEF could be removed in demagnetizing fields of 10–15 mT. The mean directions of the characteristic magnetizations (site 30: decl. = 24.7, incl. = −16.9 and site 31: decl. = 11.3, incl. = −19.2; Fig. 4; Table 1) are however quite different when compared to the sills from Finland (Fig. 3c) and are not combined with those to derive a mean of the complex.

4.4. The Västerbotten complex, northern Sweden

Dolerite sills in northern Sweden have been sampled at 11 sites (Fig. 1). On the basis of microscope studies the sills are classified as olivine diabases and are medium to fine grained with subophitic to ophitic texture. The magnetite has a primary character and occurs mainly in composite grains with ilmenite lamellae. Small amounts of pyrite and sometimes pyrrhotite are often observed. A characteristic magnetization is usually isolated in a narrow range of unblocking temperatures of ca 560–580°C (Fig. 2d). The coercivities of the natural remanent magnetization (NRM) vary, but are generally high, with MDF's of ca 25 mT (Fig. 2d). This suggests that the carrier of remanence in these sills is predominately pseudo-single-domain magnetite-titanomagnetite. The samples with a larger portion of low coercivity remanence carry a magnetization with a direction similar to that of the present earth field (PEF). In these samples a characteristic remanence is often isolated in alternating fields higher than 25 mT. In sills where the NRM is mainly carried by pseudo-single-domain grains there is only a small portion of soft remanence, which is erased at 5–10 mT

Table 3
Mean palaeomagnetic directions of the central Scandinavian dolerite complexes[a]

Dyke swarm	B/n	Decl.	Incl.	α_{95}	k	Plon.	Plat.	A_{95}
Northern Sweden								
(this study)	10/73	45.0	−45.0	6.9	50	−7.7	158.3	6.9
(+site 11)	11/79	41.5	−43.2	8.6	29	−5.4	161.0	8.4
Ulvö complex								
(this study)	4/37	48.1	−45.3	10.5	77	−7.5	156.5	10.6
(previous data included)	10/498	46.3	−43.5	5.8	71	−5.7	157.6	5.7
Jämtland complex								
(this study)	14/122	37.8	−37.3	9.1	20	0.6	160.5	8.2
(previous data included)	16/145	38.8	−38.6	8.1	22	−0.5	159.8	7.4
Satakunta complex								
(−sites 30 and 31; this study)	7/69	44.7	−40.3	4.5	181	−1.5	161.5	4.2
(−sites 30 and 31; previous data included)	11/169	45.7	−37.8	4.7	97	0.0	157.1	4.3
Sites 11, 30 and 31	3/25	17.0	−19.0	10.4	141	17.2	179.7	11.1
Dala complex								
(previous work)	5/53	35.1	−32.7	12.0	42	5.9	160.4	10.2

[a] *Note* For explanation see Table 1 and Table 2.

demagnetizing fields (Fig. 2d). Also in thermal demagnetizations low unblocking temperature components are isolated and from the similarity in direction with that of the PEF a viscous origin for this remanence is suggested.

The site mean directions are fairly well defined and for a majority of sites the precission parameter is higher than 30 (Table 1). Apart from the mean direction of site 11 (Table 1; Fig. 3d) the site mean directions cluster around a mean direction for these sills at 45.0° declination and −45.0° inclination (Table 3; Fig. 4). If site 11 is included the mean direction does not change significantly (decl. = 41.5°, incl. = −43.2°), although the precision of the mean decreases (Table 3).

5. Anisotropy of magnetic susceptibility

The anisotropy of magnetic susceptibility (AMS) was measured to define the magnetic fabric of the sills at 35 of the sites. Two dolerite dykes that cut sills of the Ulvö Complex (sites 14a, c; Fig. 1) were also included in the study. On average nine samples from each site were measured. Basically, the AMS of a rock is described by a second-order tensor (e.g. Hrouda, 1982), which can be visualized by an ellipsoid with three principal axes of maximum, intermediate and minimum susceptibility (k_{max}, k_{int} and k_{min}, respectively). For multi-domain magnetite the k_{max} axis is parallel to the long axis of an ellipsoidal grain and for most rocks the k_{max} corresponds to the mineral lineation, whereas the k_{min} axis corresponds to the pole of the plane of magnetic foliation (Tarling and Hrouda, 1993). The degree of anisotropy is characterized by $P = k_{max}/k_{min}$ and the shape of the anisotropy ellipsoid is defined by the shape parameter T (Jelínek, 1981). For $-1 < T < 0$ the ellipsoid is prolate (cigar-shaped) and for $0 < T < 1$ the ellipsoid is oblate (disc-shaped). If $T = 0$ the ellipsoid is neutral. The AMS and temperature variation of magnetic susceptibility of rocks were measured with a Kappabridge KLY-3 and a CS-3 instrument, respectively (Geofyzika Brno). The statistical analysis of the AMS data was performed using a technique of Jelínek (1978).

It is believed that elongated particles in a flowing magma become imbricated along the dyke margin and when the magma crystallizes the oxides will mimic the fabric of the preexisting silicate phases (Tauxe et al., 1998). If ferromagnetic grains of multi-domain size dominate the magnetic susceptibility of the rock the preferred orientation of the k_{max} axis will be parallel to the flow direction of the magma. Since Knight and Walker (1988) demonstrated the use of AMS as an indicator of direction of magma flow in dykes the method has been used in several studies to define flow fabrics (e.g. Bates and Mushayandebvu, 1995; Raposo, 1997). Dragoni et al. (1997) presented a theoretical model for the behavior of rigid ellipsoidal grains in a flowing magma driven by a pressure gradient. They compared the model results with AMS fabrics from Ferrar dolerite sills on east Antarctica and concluded that the magnetic lineation is generally a good marker of the magma flow direction. The exception was in the center of the sill body which is essentially strain-free and no preferred orientation can be defined. Large scatter or absence of preferred directions in the magnetic lineations may thus indicate that the measured samples were not collected close to the sill margin.

In our study we could only in rare cases observe the contact between the host rock and the sill and it is thus not possible to determine where in the sill body the samples were collected. On basis of statistical requirements for the confidence of mean AMS directions (Jelínek, 1978) a maximum size limit for the semi axes of the 95% confidence ellipse of the mean lineation direction is set to 25°. Sites that have mean directions with confidence ellipses above this limit are considered to poorly defined to represent a statistically reliable direction, and are therefore rejected.

5.1. Results of the AMS study

The degree of anisotropy is generally below 10% for the sills and dykes in this study (on average $P = 1.05$; Fig. 5b, c; Table 4), which is the usual for igneous rocks with a primary magnetic fabric (Hrouda, 1982). The site mean susceptibility average is 31×10^{-3} SI, ranging from 13.7 ×

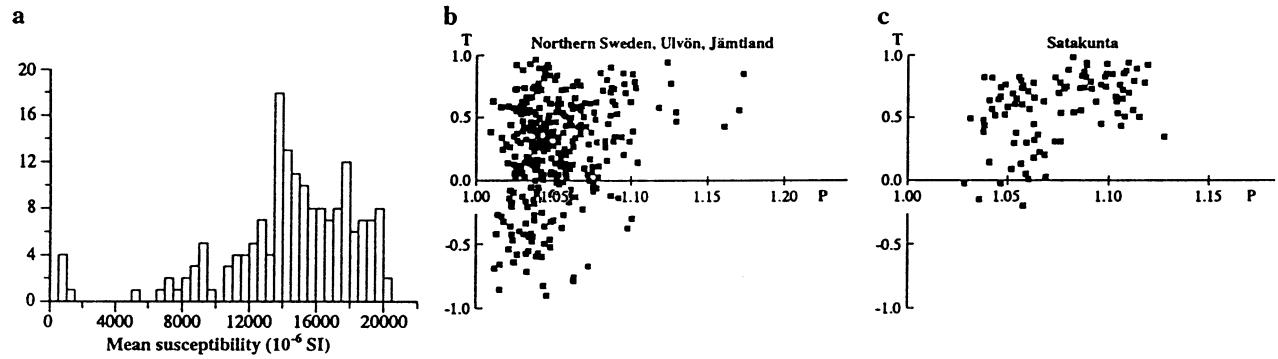


Fig. 5. Anisotropy parameters of: (a) the dolerite sills; (b) Histogram of the mean susceptibility for all specimens ($n = 434$); (b) the shape parameter T plotted as a function of the degree of anisotropy for the sills in northern Sweden, the Jämtland and the Ulvö complexes ($n = 339$); (c) the shape parameter T plotted as a function of the degree of anisotropy for the Satakunta Complex (including site 31; $n = 95$).

10^{-3} (site 19) to 80.4×10^{-3} SI (site 14a) (Fig. 5a; Table 4). Susceptibility–temperature data indicate that magnetite or titano-magnetite is the main carrier of the susceptibility of the sills. No significant differences in susceptibility or anisotropy parameters are found between the three rock-complexes in Sweden (Västebotten, Ulvön and Jämtland). A majority of the specimens have oblate shape anisotropy ellipsoids, but there are examples of specimens with strongly developed prolate shaped ellipsoids as well (Fig. 5b). At the Satakunta Complex the ellipsoids have stronger developed oblate shape and only four of the 69 samples have dominant prolate shaped ellipsoids (Fig. 5c). The average degree of lineation is slightly lower for the Satakunta Complex ($L = 1.008$) than for the Västerbotten ($L = 1.012$), Ulvön ($L = 1.012$) and Jämtland ($L = 1.011$) complexes, however, the differences are not statistically significant. The minimum anisotropy axes in the sills of all the four complexes are, with few exceptions, clustered in subvertical directions and the magnetic foliation planes have an average dip of 13°. One exception from this pattern of distribution is found at the site 26 (Fig. 1) where there is a tendency for a girdle distribution of the k_{min} axes, which may indicate that the grains underwent rotation during the magma flow. However, at site 26 the sill has a deviating remanence direction, probably indicating a post intrusion tectonic disturbance. The maximum and intermediate axes are often distributed in a subhorizontal girdle pattern, which makes it more difficult to define a reliable mean direction of the lineation (Fig. 6).

At 15 of the 35 sites the defined mean directions of lineation satisfy our statistical demands, which means that they have semi-angles of the 95% confidence ellipses that are smaller than 25° (Table 4). At site 14 in the Ulvö Complex three localities were sampled and at two of them it was possible to define a reliable direction of lineation. The difficulty of defining statistically reliable directions of lineation is to some extent caused by the limited number of samples. However, in many cases it most likely indicates a true absence of preferred direction of lineation, i.e. the samples may have been collected from the central, strain free part, of the sometimes 250 m thick sill body, where no preferred direction exists (Dragoni et al., 1997). Plotting the directions of sample lineation for each complex on an equal area net the contour plots (Fig. 7) generally indicates a girdle distribution of the lineation at the Ulvö, Jämtland and Satakunta complexes, however, with clusters of directions. In the Västerbotten Complex the directions are less scattered and the clusters are in NNW–SSE to NW–SE trending directions. These trends of the magnetic lineation in the Västerbotten sills are also expressed in the mean directions of the sites and NW–SE directions dominate. These directions roughly coincide with the NNW–SSE to NNE–SSW trending lineations

of the Ulvö complex located southeast of the Västerbotten Complex (Fig. 8). A NW–SE trend is also dominant in the Jämtland Complex further to the southwest, however, there is one site with a mean lineation directed NNE–SSW. In the Satakunta Complex in Finland, the situation is slightly different. Significant mean directions of lineation can be determined in four of the seven sites. In three of these sites the lineations trend in NNE–SSW directions, while in the fourth site the lineation trend is NW–SE (Fig. 8).

We interpret the AMS data as a primary flow fabric indicator of the rocks at all 35 sites. A flow origin is supported by AMS data from a sill in the

Table 4
Magnetic anisotropy parameters for the Central Scandinavian Dolerite Group (CSDG)[a]

Site	*N*	*K* mean	*P*	*L*	*F*	Decl. *K*1	Incl. *K*1	α_{95} K1
1	5	43 400	1.054	1.019	1.034	130	1	9
2	8	38 000	1.048	1.022	1.026	136	10	21
3	5	33 300	1.045	1.019	1.026	313	1	41
4	9	40 200	1.020	1.004	1.016	269	0	38
5	6	39 500	1.021	1.003	1.018	356	5	68
6	14	28 000	1.031	1.012	1.018	139	5	22
7	8	16 000	1.029	1.006	1.024	125	4	39
8	7	15 800	1.047	1.006	1.040	326	0	25
9	7	26 000	1.032	1.016	1.015	163	7	17
10	5	34 800	1.050	1.009	1.041	160	8	39
12	10	26 200	1.057	1.009	1.048	127	6	28
13	9	15 000	1.027	1.005	1.021	344	1	28
14a	6	80 400	1.131	1.023	1.106	161	10	23
14a dyke	6	23 900	1.059	1.027	1.031	113	5	18
14b	8	68 100	1.096	1.012	1.084	203	6	18
14c	8	61 800	1.039	1.010	1.029	100	14	52
14c dyke	7	20 100	1.025	1.012	1.013	359	8	20
15	11	14 300	1.050	1.010	1.040	145	4	41
16	13	15 800	1.031	1.007	1.023	219	7	27
17	8	17 700	1.059	1.025	1.033	191	8	11
19	7	13 700	1.028	1.004	1.024	344	1	63
20	21	39 900	1.050	1.007	1.043	71	4	32
21	11	60 500	1.032	1.010	1.022	314	0	20
22	9	15 200	1.044	1.004	1.040	313	8	72
23	11	18 300	1.052	1.023	1.028	154	16	23
24	9	16 000	1.034	1.008	1.026	43	13	33
25	9	38 900	1.041	1.017	1.024	158	9	28
26	9	20 500	1.029	1.016	1.013	274	2	12
27	9	20 900	1.028	1.003	1.026	358	1	66
28	7	60 400	1.030	1.003	1.027	355	8	29
29	10	27 700	1.029	1.017	1.012	297	2	13
31	10	15 500	1.047	1.006	1.065	39	3	32
32	10	26 800	1.049	1.004	1.045	331	2	37
33	10	29 800	1.105	1.008	1.096	302	10	16
34	10	27 800	1.087	1.007	1.079	16	6	20
35	10	27 800	1.041	1.005	1.036	211	4	38
36	12	18 600	1.077	1.008	1.068	8	4	21
37	10	28 400	1.090	1.006	1.084	183	7	30
38	8	25 900	1.092	1.017	1.074	16	2	25

[a] *N* is the number of samples, *K* mean is the mean susceptibility (10^{-6} SI), *P* is the degree of anisotropy, *L* is the degree of lineation, *F* is the degree of foliation, Decl. *K*1 is the declination of the magnetic lineation, Incl. *K*1 is the inclination of the magnetic lineation and α_{95} *K*1 is the largest semi-axes of the confidence ellipse about the *K*1 axis.

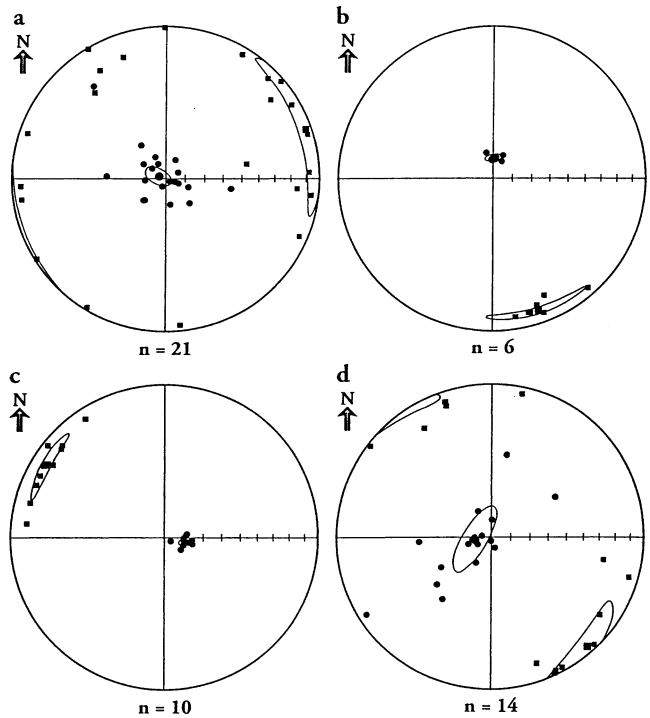


Fig. 6. Examples of directions of the maximum (filled squares) and minimum (dots) anisotropy axes of susceptibility in sills from: (a) the Jämtland Complex; (b) the Ulvö Complex; (c) the Satakunta Complex; and (d) the northern Sweden sills. The directions are shown in downward equal area projection and the ellipses are 95% confidence limits for the directions of the susceptibility axes.

Jämtland complex (site 20) that cuts a deformation zone. While the host rock is deformed and has a susceptibility anisotropy with a vertical foliation plane corresponding to the deformation-related mineral fabric, the anisotropy of the sill exhibits a magnetic foliation plane parallel to the plane of the sill. A flow origin is also supported by the preferred orientation of plagioclase crystals in the layered sill at Ulvön.

The two dykes at Ulvön (sites 14a and c) are subvertical and strike WNW (site 14a) and north (site 14c). It is suggested that both dykes carry a primary AMS fabric since the degree of anisotropy is less than 10%. The magnetic lineation of the dykes clusters subhorizontally, which indicates that the dykes were fed by horizontal fluxes and that Ulvön was situated far from the magma source at the time of emplacement (Raposo, 1997). The inferred flow direction of the sill at site 14a is NNW–SSE and ca 45° away from the trend of the cutting dyke. This may indicate that the two rocks were emplaced at different times and under different stress conditions.

6. Geochemistry

In 32 samples from Västerbotten and 13 from Jämtland, major and trace elements were analysed by SGAB Analytica in Luleå by using ICP-AES. Apart from the new chemical analyses, there are previously presented analyses from the Jämtland area (Gorbatschev et al., 1979), from the Ulvö area (Lundqvist and Samuelsson, 1973; Larsson, 1980; Thomas Lundqvist, 2000, unpublished analyses) and from the Satakunta area (Hämäläinen, 1987).

The CSDG rocks are not much altered, and have largely kept their primary mineralogy. The basic intrusions in Västerbotten, Jämtland, Ulvön and Satakunta areas all have similar tholeitic compositions, which indicate an intraplate origin.

7. Discussion

The mean remanence directions (Figs. 3 and 4) and the calculated palaeomagnetic pole positions (Fig. 9a and b; Table 3) of the sills from the Ulvö, the Jämtland and the Satakunta complexes in this study are not significantly different. The same is true when our new data are combined with old data from these complexes and compared also with the Dala Complex (Tables 2 and 3). The similarity in palaeomagnetic directions between these complexes was also observed by Bylund and Pesonen (1987), however their observations were based on a smaller amount of data, especially in the case of the Jämtland Complex. The new palaeomagnetic data from the northern part of Sweden, the Västerbotten Complex (site 1–11), fall into the same pattern and the mean direction and mean pole for these sills (site 11 not included) are not significantly different from those of the other complexes (Figs. 4 and 9b). This means that the Central Scandinavian Dolerites are not restricted to the central part of the shield but have intruded the crust as far north as 65.7° latitude. Up to now we have not identified dolerites of this magnetization age north of the Skellefte district, which marks a boundary between the northern and central Svecofennian subprovinces (Fig. 1). The similarity in pole positions also means that the sills of the different complexes were most likely intruded within a quite narrow time interval, unless there was little or no polar wander at this time. However, the polarity of the remanence is the same for all complexes, which argue for a narrow time interval of the intrusions.

The mean remanence directions from three sills (sites 11, 30 and 31) form a group of similar directions with a mean (decl. = 17.0°, incl. = −

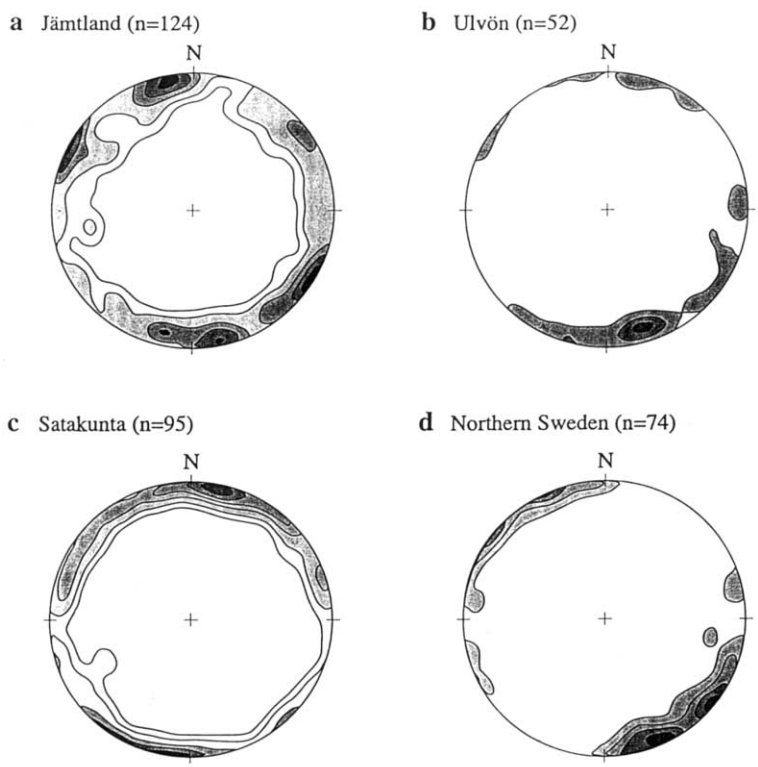


Fig. 7. An equal area plot of the directions of magnetic lineation (maximum axis) in dolerite samples from: (a) the Jämtland Complex; (b) the Ulvö Complex; (c) the Satakunta Complex; and (d) northern Sweden sills (the Västerbotten Complex). The directions are shown in a downward projection.

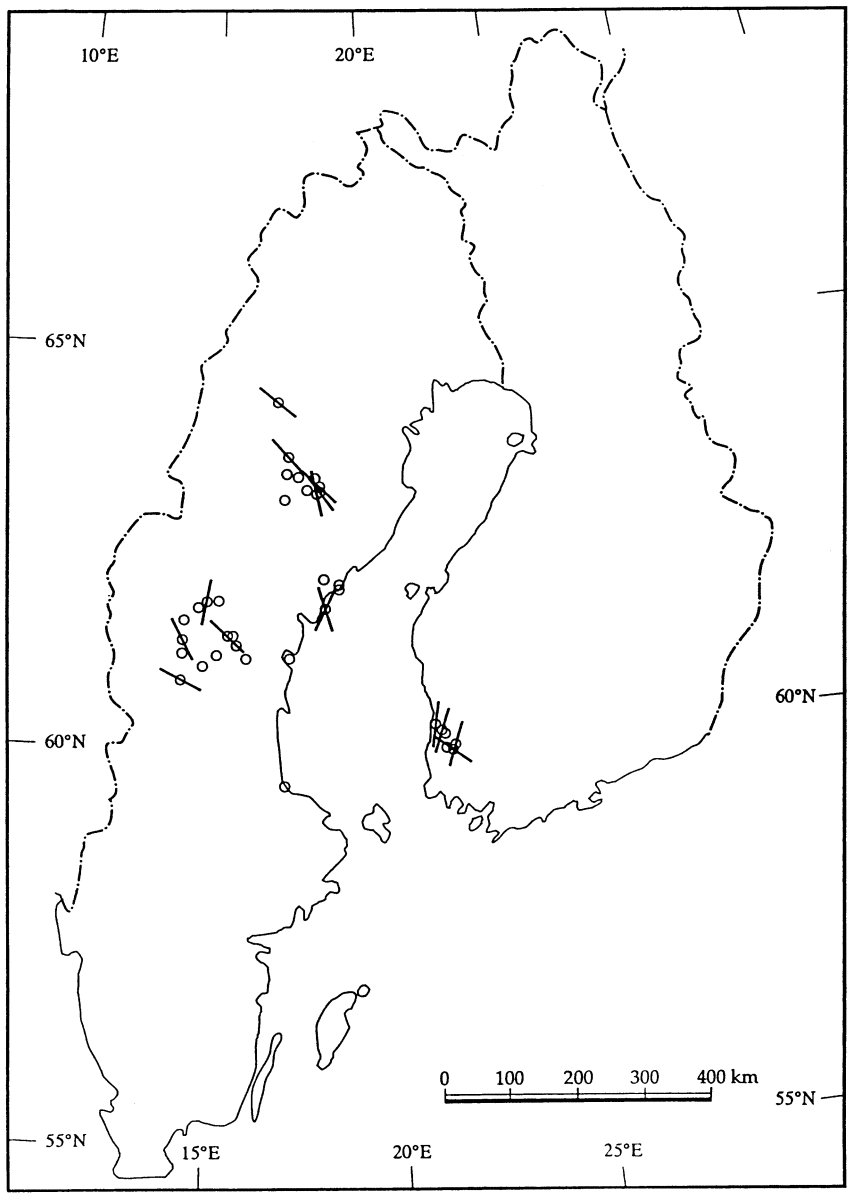


Fig. 8. Site mean strike directions of the magnetic lineation from 15 sites, where the directions could be defined with a radius of the cone of confidence less than 25°. The dip of the lineation is most often close to zero and is not accounted for in the figure.

19.0°, Table 3) significantly different from the mean directions of all other sills. The geochemical data show a significantly different composition for the sill at site 11, which means that the difference in remanence directions is probably due to a different origin and age of these dolerites. The calculated pole (Plat. = 17.2°, Plon. = 179.7°, Table 3, Fig. 9b) coincides with the ca 1.3 Ga reference pole for the Fennoscandian Shield (Elming et al., 1993), and indicates that they may be older than the CSDG.

The pole calculated from the sill at site 26 in the Jämtland Complex is located at an anomalously high southern latitude (Fig. 9a). This sill also has

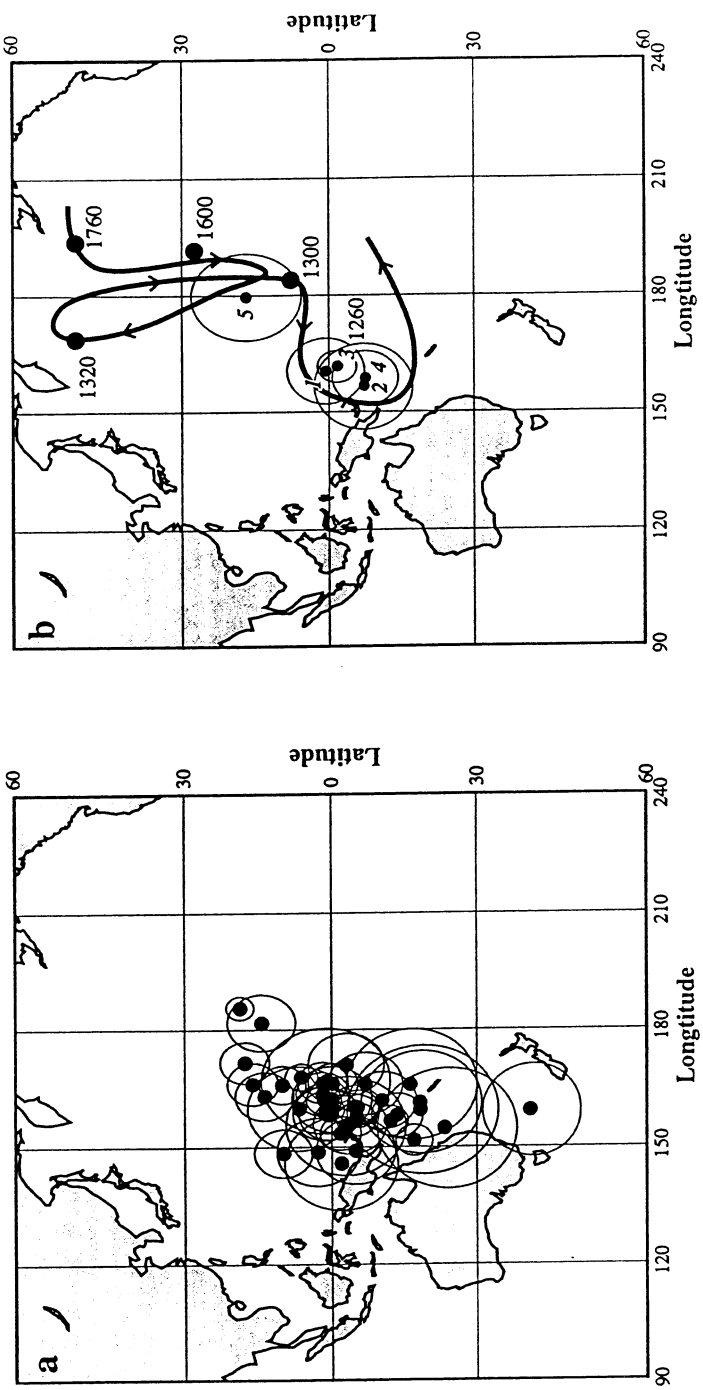


Fig. 9. (a) Site mean virtual geomagnetic pole (VGP) positions plotted together with A_{95}, the half-angle of the 95% circle of confidence. The pole at a high southern latitude is calculated from the anomalous site 26; (b) Mean VGP position accompanied by A_{95} for the dolerites of: (1) the Jämtland Complex; (2) the Ulvö Complex; (3) the Satakunta Complex; (4) the dolerites from northern Sweden, the Västerbotten Complex; and for the anomalous dolerites at sites 11, 30 and 31(5).

anomalous AMS directions, which may suggest that it is tectonically disturbed.

The magnetic fabric data indicate a fairly uniform ca NW or SE directed magma flow of the dolerite sills in northern and central Sweden (the Västerbotten, Jämtland and Ulvö complexes, see Fig. 8). These sites also have similar signatures of the anisotropy parameters, which may indicate that the magma flows were formed under similar physical conditions. The CSDG rocks have intruded into a much older bedrock, and the geochemical data indicate a clear within-plate setting. The similarity in geochemical composition, also suggests that the magma came from the same source. This source may have been located in the central part of the shield or N–NWof the Västerbotten Complex as indicated from the magnetic lineations. The horizontal lineation in the dykes of the Ulvö Complex suggests that the source of magma was located far from these dykes.

The mean trend of the magnetic lineation of the Satakunta Complex (Finland) indicates a magma flow in predominantly a NNE–SSW direction (Fig. 8). This may suggest that the magma of the Satakunta area comes from a different source when compared with the other complexes, however, their chemical compositions are similar.

AMS measurements were previously performed on the ca 1.25 Ga old Midsommersö dolerite sills and the Zig–Zag Basalt in northeastern Greenland (Marcussen and Abrahamsen, 1983). The basalt is a tholeitic flood basalt and the dolerites have a quartz-tholeitic composition. Kalsbeek and Jepsen (1983) reported some major element analyses of the Midsommersö dolerites. Except for fractionation effects, these dolerites show a wide variation in chemical composition due to both hydrothermal alteration and assimilation of the wall rock. This makes a detailed comparison of the composition of these dolerites with those from the Baltic Shield difficult. At three sites in the basalt well defined susceptibility axes were determined. The minimum axes of anisotropy are generally vertical for the basalt, indicating horizontal flow, and the maximum axes are in a NE–SW direction, which is interpreted to reflect the flow direction of the magma. At two localities in the Midsommersö dolerite (locality 3; Marcussen and Abrahamsen, 1983) consistent AMS results were obtained. In one feeder dyke the mean maximum AMS-axis trends in a NE–SW direction (215° E) and the flow plane is dipping ca 42° to the southwest and parallel to the dip of the dyke (40° SW). In another, almost horizontal (5–10° SW dip) dyke, the maximum anisotropy axes are also directed NE–SW (229°). According to the authors these results are in excellent agreement with geological evidence for the flow direction and the dip of the flow plane. In three other sites from a major sill well defined anisotropy axes with almost vertical maximum axes were defined, which indicate a vertical flow.



Fig. 10. Tectonic reconstruction of Laurentia and Baltica at 1.27 Ga based on palaeomagnetic data and geological criteria (Buchan et al., 2000). The arrows denote the flow direction of magma in the sills in Fennoscandia (fulldrawn line) and Greenland (hatched line) as indicated from AMS and geological data. Note that the flows are more or less parallel and perpendicular to the contact between Fennoscandia and Greenland.

On basis of key palaeomagnetic poles for Laurentia and Baltica Buchan et al. (2000) presented a possible location of northern Baltica adjacent to northeastern Greenland at ca 1.27 Ga (Fig. 10). In this reconstruction the 1.8–1.5 Ga old Gothian and Labradorian belts are aligned and Laurentia and Baltica may have been joined in this time

period. The key pole for Baltica is based on palaeomagnetic data from the Central Scandinavian Dolerites and this key pole (Plat. = 4°, Plon. = 158°, $A_{95} = 4°$) is not significantly different from the mean pole of the sills in our study (Plat. = −3.8°, Plon. = 159.3°, $A_{95} = 5.6$; Table 3). Laurentian key poles of corresponding age are defined from the Mackenzie and Sudbury dykes, while there are no key poles of this age from Greenland. However, when the position of Greenland is reconstructed to northwestern North America using a Mesozoic fit (Roest and Srivistava, 1989) the pole of the ca 1.25 Ga Gardar dykes falls near the 1.267 Ga Mackenzie and the 1.235 Ga Sudbury poles. In this reconstruction the poles of the Midsommerö dolerites and the Zig–Zag basalts are also similar to those of the Laurentian key poles (Buchan et al., 2000). The polarity of remanences in the Midsommerö dolerites and the Zig–Zag basalts are the same but opposite to what has been defined for dykes of similar age in central North Greenland (Abrahamsen and Van der Voo, 1987). Poles calculated for the ca 1.15 Ga old Late Gardar dykes fall in a similar way close to the poles of the 1.14 Ga Abitibi dykes and the Logan sill for Laurentia, which suggests that Greenland formed a part of Laurentia at that time.

To fit the reconstruction of Buchan et al. (2000) Laurentia has to be rotated clockwise and Baltica counterclockwise relative to their present positions (Fig. 10). In that reconstruction the polarity of remanences of the Midsommerö dolerites and the Zig–Zag basalts is opposite to that of the CSDG in Fennoscandia. The reconstruction suggests that northeastern Greenland may have been attached to the northwestern part of Baltica not far from where the Västerbotten sills crop out. The predominantly NW–SE lineation of the magnetic susceptibility in the Swedish sill complexes will in this reconstruction be more or less parallel to the corresponding lineation and proposed magma flow in the sills of similar age in northeastern Greenland. Marcussen and Abrahamsen (1983) suggested, on basis of geological and AMS data, that the flow of the magma in Greenland may have been from the NE to the SW and in one sill vertical, that is a flow from the coast towards the interior of Greenland.

This means that the magma source should have been located close to the coast or just outside present day Greenland. Such a location of the source also fits into the flow pattern in Baltica on the tectonic reconstruction (Fig. 10). The parallel flow directions in the sills in northwestern and western Baltica and the basalt and sills in northeastern Greenland suggest a common magma source. The sills in Greenland and those in Baltica are of tholeitic composition indicating an intraplate origin, and support the interpretation of joined continents at that time. The location of a magma source in the reconstructed contact between Baltica and Laurentia may be a sign of rifling that reflects the break up of the Baltica/Laurentia supercontinent. The tensional regime that is indicated by the huge sill complexes in Baltica would then be a result of this plate tectonic event.

8. Conclusion

The Central Scandinavian Dolerites represented by the Västerbotten, Ulvö, Jämtland, Satakunta and Dalarna complexes cover more than 100 000 km^2 in the Fennoscandian Shield. Geochemically the dolerites are rather uniform and have a composition typical of mantle derived melts formed in continental tensional settings. Palaeomagnetic data suggest a small time difference between the intrusion of the dolerites. In a palaeomagnetic reconstruction of Baltic a versus Laurentia at ca 1.27 Ga the two continents were joined to each other, with NE Greenland attached to NW Baltica. Measurements of the anisotropy of magnetic susceptibility show a magnetic fabric that indicates a fairly uniform ca NW or SE directed magma flow in the dolerites in northern and central Sweden. Corresponding measurements on basalts and dolerites in NE Greenland indicate a SW flow direction, which in the tectonic reconstruction of Laurentia and Baltica is parallel to the flow direction of the sills in Sweden. This may suggest a common magma source located in the reconstructed contact between the two continents. Both the sills in Greenland and those in Sweden are of tholeitic composition indicating an in-

traplate origin, which supports the interpretation of joined continents at that time. The tensional regime, that is indicated by the huge sill complexes, reflects, in our interpretation, the break up of Baltica from Laurentia at ca 1.27 Ga ago.

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